

BRAUNSCHWEIG, Germany

WMO: 103480

Lat: 52.2914N

Lon: 10.4464E

Elev: 269

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	14.5	5.8	7.5	12.0	10.2	9.4	16.4	25.0	44.1	22.7	41.3	4.8	90	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	83.6	66.2	80.0	64.6	76.8	63.1	67.9	79.1	66.1	77.0	64.5	74.1	8.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.8	71.8	62.0	84.0	70.4	60.5	79.4	68.9	32.4	79.2	30.9	77.1	29.7	74.3	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.5	16.5	DB	6.7	90.5	10.5	4.3	-0.8	93.6	-7.0	96.1	-12.8	98.5	-20.5	101.6	
			WB	6.3	71.8	10.4	1.6	-1.2	73.0	-7.2	74.0	-13.0	74.9	-20.6	76.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	33.6	32.9	39.7	46.6	54.8	60.2	63.8	64.0	56.7	49.6	41.8	36.8	(6)
(7)		DBStd	12.99	10.13	9.40	7.38	6.70	6.44	6.05	5.47	5.24	4.40	6.56	6.53	7.17	(7)
(8)		HDD50	2244	509	482	325	142	26	2	0	0	3	89	254	412	(8)
(9)		HDD65	6176	974	900	784	553	320	169	90	83	252	479	697	875	(9)
(10)		CDD50	1681	0	2	6	39	176	307	428	435	203	76	7	2	(10)
(11)		CDD65	138	0	0	0	0	5	24	53	53	2	1	0	0	(11)
(12)		CDH74	1339	0	0	0	2	70	245	508	483	27	4	0	0	(12)
(13)		CDH80	296	0	0	0	0	3	52	111	129	1	0	0	0	(13)
(14)	Wind	WSAvg	7.8	9.8	8.6	8.7	7.6	7.2	6.9	6.7	6.5	7.4	7.5	8.3	8.6	(14)
(15)	Precipitation	PrecAvg	26.4	2.3	1.7	1.8	1.7	2.3	2.2	3.0	2.8	2.2	2.1	2.1	2.2	(15)
(16)		PrecMax	36.7	4.3	3.5	4.4	3.3	7.4	3.7	7.6	6.9	5.6	5.8	4.3	5.1	(16)
(17)		PrecMin	20.2	0.4	0.5	0.4	0.3	0.7	0.7	1.0	0.7	0.4	0.6	0.0	0.7	(17)
(18)		PrecStd	4.1	1.1	0.9	1.1	0.7	1.5	0.9	1.5	1.3	1.3	1.1	1.1	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	58.3	63.2	71.2	79.8	86.3	86.2	88.0	78.8	72.9	58.4	54.9	(19)
(20)			MCWB	48.6	51.1	51.2	56.1	61.2	66.3	67.0	67.8	63.4	62.1	51.4	49.4	(20)
(21)		2%	DB	50.1	52.8	56.4	66.5	75.7	80.2	83.3	83.5	72.7	66.3	55.4	52.7	(21)
(22)			MCWB	46.8	46.9	49.0	53.7	59.6	65.0	66.8	65.5	61.4	57.8	50.7	48.4	(22)
(23)		5%	DB	47.8	47.6	52.9	62.4	71.6	76.3	79.9	79.2	68.4	61.9	53.1	49.9	(23)
(24)			MCWB	44.4	43.4	46.9	51.3	58.2	62.7	65.3	64.3	59.3	55.9	49.4	46.8	(24)
(25)		10%	DB	45.1	44.3	50.1	58.5	67.7	72.3	76.3	75.4	65.4	59.1	51.0	46.9	(25)
(26)			MCWB	42.0	41.0	45.4	49.5	56.8	60.9	64.0	63.0	57.6	54.1	48.0	44.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	51.4	51.8	57.1	63.6	69.1	70.7	69.2	65.5	62.7	53.3	50.3	(27)
(28)			MCDWB	52.3	57.6	60.7	67.5	74.9	80.3	82.7	83.8	73.6	70.1	56.5	53.3	(28)
(29)		2%	WB	47.2	47.5	49.9	54.5	61.2	66.2	68.4	67.0	62.5	59.3	51.4	48.9	(29)
(30)			MCDWB	50.0	52.1	55.3	64.0	72.2	77.9	79.3	78.3	69.3	65.1	54.2	51.9	(30)
(31)		5%	WB	44.4	43.7	47.8	52.4	59.3	64.0	66.5	65.6	60.7	56.6	49.8	47.1	(31)
(32)			MCDWB	47.5	47.1	51.9	61.1	69.9	73.6	77.4	75.9	67.5	60.5	52.6	50.1	(32)
(33)		10%	WB	42.1	41.2	45.9	50.2	57.3	62.0	64.6	64.1	58.7	54.8	48.0	44.5	(33)
(34)			MCDWB	45.0	44.2	49.9	57.5	66.0	70.4	74.4	72.9	64.3	58.5	51.1	46.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.9	11.4	15.4	17.0	16.0	17.1	17.1	14.0	11.9	9.2	7.3	(35)
(36)			MCDBR	9.5	13.5	16.7	22.3	23.6	23.6	24.2	24.5	20.9	16.6	12.8	10.3	(36)
(37)			MCWBR	7.6	9.5	10.2	12.2	11.8	11.2	11.2	10.8	11.5	10.2	9.2	8.3	(37)
(38)		5% WB	MCDBR	9.1	12.5	14.7	19.6	21.0	20.5	21.5	20.0	17.6	15.1	11.6	9.8	(38)
(39)			MCWBR	7.9	9.4	9.6	11.4	11.6	10.9	11.4	9.8	10.7	9.9	9.0	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.339	0.375	0.411	0.414	0.411	0.428	0.420	0.385	0.358	0.337	0.312	(40)	
(41)		taud	2.364	2.347	2.264	2.212	2.242	2.270	2.232	2.268	2.357	2.417	2.399	2.362	(41)	
(42)		Ebn at Noon	215	244	256	260	266	267	260	255	251	234	205	194	(42)	
(43)		Edh at Noon	19	26	34	40	41	40	41	38	31	24	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	212	434	773	1266	1556	1679	1577	1337	980	557	252	162	(44)	
(45)		RadStd	22	60	78	150	178	134	216	113	115	58	28	14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page